

INIYAN ANDREWS JOSEPH

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RESEARCH INTERESTS

Deep Reinforcement Learning (PPO/DQN) for Stochastic Control, Generative World Models for Financial Time-Series, Limit Order Book (LOB) Microstructure Modeling, and Trustworthy AI in Algorithmic Trading.

EDUCATION

National Institute of Electronics & Information Technology (NIELIT)

Calicut, India

M.Sc. in Computer Science (Specialization in Data Science)

Expected May 2026

- **CGPA:** 8.55/10.0
- **Master's Thesis (*Bithax*):** Deep Reinforcement Learning for High-Frequency Digital Asset Trading – Engineered a Proximal Policy Optimization (PPO) agent to navigate a 42-dimensional non-stationary state-space, mathematically modeling execution friction as a Markov Decision Process penalty.

The American College

Madurai, India

B.Sc. in Computer Science

2021 – 2024

- **CGPA:** 7.4/10.0

QUANTITATIVE RESEARCH & PROJECTS

Deep RL in Limit Order Books (Thermodynamic Decay)

Jan 2026 – Present

Independent Researcher | Python, PyTorch, OpenAI Gym, Stable-Baselines3

- **Methodology:** Engineered an execution-penalized Markov Decision Process (MDP) for high-frequency Limit Order Books (LOB). Enforced strict state-space stationarity using Augmented Dickey-Fuller (ADF) tests and Fractional Spread dynamics.
- **Validation:** Implemented Combinatorial Purged Walk-Forward Optimization (WFO) to definitively eliminate temporal data leakage.
- **Results:** Achieved an Out-Of-Sample **Deflated Sharpe Ratio** > 2.0 , passing the 95% significance threshold by mathematically rejecting Multiple Testing Bias through Bailey-Lopez de Prado approximations.

High-Frequency Bitcoin Trading using Deep RL (Project Bithax)

2025

Master's Thesis Research | Python, PyTorch, PPO

- **Methodology:** Formulated the BTC/USD market as a continuous state-space MDP with a 42-dimensional feature vector, directly incorporating 0.05% taker fees into the reward function to penalize latency/friction.
- **Results:** Achieved a **Sharpe Ratio of 1.9** and a **Profit Factor of 1.9** under strict drawdown constraints, significantly outperforming Buy-and-Hold heuristics during non-stationary regime shifts.

Institutional Tail-Risk Modeling (Quant Risk Modeling)

2025

Independent Researcher | Python, Scipy, Monte Carlo

- **Methodology:** Rejected retail point-estimate backtesting. Built a 50,000-path Monte Carlo bootstrap engine to map 95% Confidence Intervals for strategy ruin probability.
- **Results:** Mathematically isolated catastrophic left-tail vulnerability and engineered a simulated Gamma Scalping protocol to strictly bound the portfolio ruin probability to $< 0.1\%$.

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, Stable-Baselines3, OpenAI Gym, Transformers, LLMs (Llama3).

Finance: Quantitative Risk Modeling (VaR, Monte Carlo, DSR), Algorithmic Trading (HFT, Maker-Taker Models), Time-Series Stationarity Analysis (ADF, CPCV).

Tools: Python (Advanced), C++, SQL, Git, Docker, L^AT_EX, Linux.